

5. DEVELOPMENTS IN CYCLICALLY-ADJUSTED BUDGET BALANCE MEASUREMENT

5.2. THE PRO-CYCLICAL NATURE OF DISCRETIONARY MEASURES AFFECTING TAX REVENUES

The data collected by the Commission services on the impact of discretionary measures on tax

revenues in the context of the Output Gap Working Group can be used to better understand recent tax policy changes (see European Commission, 2009 for a description of the database). Before 2008 many EU countries benefited from substantial tax revenue windfalls and relatively high apparent tax elasticities. This led some to introduce generous

tax breaks which, with hindsight, proved to be unaffordable in the aftermath of the 2008/2009 global financial crisis. Graphs II.5.2-II.5.5 provide an illustration of the recent evolution of apparent tax elasticities and the output gap, with the latter measuring the business cyclical position for a limited set of countries. ⁽¹⁾ Total tax revenues and apparent tax elasticities, both gross and net of the effect of discretionary measures for each broad tax category are reported for selected countries. The net elasticities are derived using the proportional adjustment method described in European Commission (2009). For each tax category, the tax base used is nominal GDP in order to ensure direct comparability with the OECD/Commission tax elasticities used to calculate the cyclically adjusted balance, as depicted in Graphs II.5.2-II.5.5. ⁽²⁾ Apparent tax elasticities appear to be very volatile in the short-run and can sometimes depart substantially from the OECD/Commission benchmark. In certain cases, however, the impact of discretionary measures on the tax elasticity is large, yielding some substantial discrepancies between net and gross elasticities in these cases. Graph II.5.2 depicts the results for the apparent elasticity of direct taxes with respect to GDP in selected countries. Both net and gross elasticities appear to be very volatile and tend to fluctuate around the OECD/Commission benchmark elasticities, reflecting general business cycle variations as shown by the output gap values. Graph II.5.3 shows the results of an equivalent exercise for a different set of countries and indirect taxes. Again, high volatility of apparent tax elasticities can be seen with some significant departures from the OECD/European Commission benchmark also being due to the overall output variations as indicated by the values of the output gap. Graph II.5.4 which shows tax elasticities of Social Security Contributions shows lower volatility for both net and gross elasticities and a smaller impact of discretionary measures on the tax elasticities. Graph 4 plots gross and net apparent elasticities of the total tax revenues for a larger sample of countries and also shows that, while gross and net tax elasticities are often highly

correlated, the incidence of discretionary measures can in some years be rather large and drive to sizeable discrepancy between the two series. ⁽³⁾

Graphs II.5.2-II.5.5 can be used to illustrate the pro or counter-cyclical nature of discretionary measures. For instance, for a given level of output growth, tax cuts will result in the net tax elasticity being higher than the gross elasticity, as the change in tax revenues in the gross case includes the tax cut while the change in tax revenues in the net case does not. Similarly, in case of a tax increase, the net tax elasticity will be lower than the gross tax elasticity. ⁽⁴⁾ Graph II.5.4 concerning total taxes shows that discretionary measures tended to be counter-cyclical in Italy in 2003, counter-cyclical in France in 2004-2005, and in the Czech republic in 1998-2003, while they tended to be pro-cyclical in Malta in 2004-2005. ⁽⁵⁾

⁽³⁾ The correction of tax revenues taking into account the entire series of discretionary measures yields tax elasticities net of discretionary measures which can depart significantly from the gross tax elasticities series as shown by the Czech case. This reflects the fact that discretionary measures can have long-lasting effects and thus can affect tax elasticities over long periods of time.

⁽⁴⁾ Apparent net tax elasticities are higher than the gross tax elasticities when discretionary measures imply a negative variation of tax revenues (i.e. a tax cut) and lower when discretionary measures represent a tax revenue increase. To see this, one can consider the simple case where the change in tax revenues is observed over a two years period, between t and $t-1$, such that net tax elasticities in year $t-1$ is $At-1$. It can be written as follows using the proportional adjustment method, assuming that the year t represents the base year:

$$A_{t-1} = T_{t-1} \frac{T_t}{T_t - DM_t}$$

. T_t is the gross tax level and DM_t is the discretionary measure in year t (i.e. the base year). The variation in tax revenues net of discretionary measures between $t-1$ and t will be larger than the variation of gross tax revenues,

i.e., $\Delta A_t > \Delta T_t$ if: $T_t - T_{t-1} \left(\frac{T_t}{T_t - DM_t} \right) > T_t - T_{t-1}$, i.e., if $DM_t < 0$ Conversely, $\Delta A_t < \Delta T_t$ if $DM_t > 0$.

⁽⁵⁾ Following the explanation given above, during expansionary phases of the cycle, pro-cyclical discretionary measures would yield negative tax revenue variations and thus higher net tax elasticities than in a no-policy change scenario while during slowdowns discretionary measures would yield positive tax revenue variations compared to a no-policy change scenario, thus resulting in lower net tax elasticities. In both cases net tax elasticities changes would amplify variations in gross tax elasticities and this would, in principle, yield larger variance.

⁽¹⁾ Apparent tax elasticities have therefore been computed by dividing the annual growth of the revenue series (both gross and net) with the nominal GDP annual growth rate.

⁽²⁾ The OECD/Commission tax elasticities are available in European Commission (2006), Public Finances in EMU 2006, Directorate General for Economic and Financial Affairs, European Commission.

6.1. INTRODUCTION

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